The Weak Nuclear Force, Chiral Phase Leakage, and Beta Decay: Geometric Synthesis of the Four Fundamental Interactions

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Abstract

In this study, we model the weak nuclear interaction, chiral parity violation, and intermediate vector boson ($W^{\pm}, Z^0$) mass scales within the framework of Scale-Time Calculus and saturated vacuum vortex geometry. In the Standard Model, the weak force is governed by the spontaneously broken Glashow–Weinberg–Salam $SU(2)_L \times U(1)_Y$ electroweak gauge symmetry. Here, we establish this dynamics analytically as **Chiral Scale Leakage** across the hadronic vortex boundary via quantum tunneling through the vacuum saturation barrier.

Under continuous contraction, the internal quantum circulation of a saturated hadronic vortex experiences a slight chiral shear at the Planckian tension boundary. This phase leakage excites local massive vector states ($W^{\pm}, Z^0$) on the static vacuum continuum. The extremely short range ($r_W \approx 2.45 \times 10^{-18}$ m) and apparent weakness of the interaction originate from the high energy threshold required to tunnel through this saturation barrier (yielding the effective Fermi coupling $G_F \sim 1/M_W^2$). The strict left-handedness of neutrinos, free neutron lifetime ($\approx$ 879 s), in-nucleus stability via collective phase-pinning, and PMNS neutrino mixing are shown to be analytically consistent with vortex phase relaxation. Finally, the four fundamental interaction potentials are comparatively classified within a unified dimensional master flux formulation.

Keywords: Weak Nuclear Force, Scale-Time Calculus, Chiral Phase Leakage, Beta Decay, Vacuum Saturation Barrier, W/Z Bosons, Fermi Coupling.

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1 Introduction and Gauge Properties of the Weak Interaction

The weak nuclear interaction exhibits unique chiral and massive properties within the Standard Model gauge framework:

1. **Maximal Chirality / Parity Violation:** Weak charged currents couple exclusively to left-handed fermions and right-handed antifermions ($V-A$ structure).
2. **Heavy Vector Bosons:** Unlike the massless photon and gluon, electroweak force carriers ($W^{\pm}, Z^0$) possess massive rest scales ($M_W \approx 80.4$ GeV, $M_Z \approx 91.2$ GeV).
3. **Short Interaction Range:** Governed by Yukawa exponential suppression, the interaction range is sub-hadronic ($r_W \sim \hbar/M_W c \approx 2.45 \times 10^{-18}$ m).

In the Standard Model, the weak interaction is formulated through the Glashow–Weinberg–Salam $SU(2)_L \times U(1)_Y$ electroweak gauge symmetry [7, 8], building upon Fermi's original four-fermion contact theory [6]. In Scale-Time Calculus [1], this process is formulated as a **chiral phase tunneling** leaking from a saturated hadronic flux tube into the quiescent vacuum continuum.

2 Vortex Shear and Chiral Phase Leakage

As modeled in Paper 6 [5], vacuum inflow at the hadronic scale collapses into a 1D saturated flux tube. A neutron (udd) is a triple quark vortex configuration.

Throughout the contraction process, the internal angular momentum and phase circulation of this triple vortex undergo a subtle chiral shear at the Planckian saturation threshold of the static background vacuum:

$$\Delta\Phi_{\text{chiral}} = \oint \mathbb{D}_t \mathbf{v}_{\text{vortex}} \cdot d\mathbf{l} \neq 0. \tag{1}$$

This chiral shear ruptures the flux line of one of the down quarks, releasing a topological phase stress packet (W^- boson):

$$d \to u + W^- \to u + e^- + \bar{\nu}_e. \tag{2}$$

3 Fermi Constant and Geometric Derivation of W/Z Masses

The massive rest scale of the W and Z bosons ($M_W \approx 80.4$ GeV) corresponds to the **minimum phase stress threshold required to tunnel through the saturated vacuum wall**.

Theorem 3.1 (Barrier Thickness and Weak Interaction Range)**.** *The characteristic tunneling distance across the vacuum saturation barrier is:*

$$r_W = \frac{\hbar}{M_W c} \approx 2.45 \times 10^{-18}\ m. \tag{3}$$

In the low-energy regime ($E \ll M_W c^2$), the leakage cannot surmount the barrier and decays exponentially in Yukawa form. The effective Fermi coupling constant is:

$$G_F = \frac{\sqrt{2}}{8} \frac{g_W^2}{M_W^2 c^4} \approx 1.16637 \times 10^{-5}\ GeV^{-2}. \tag{4}$$

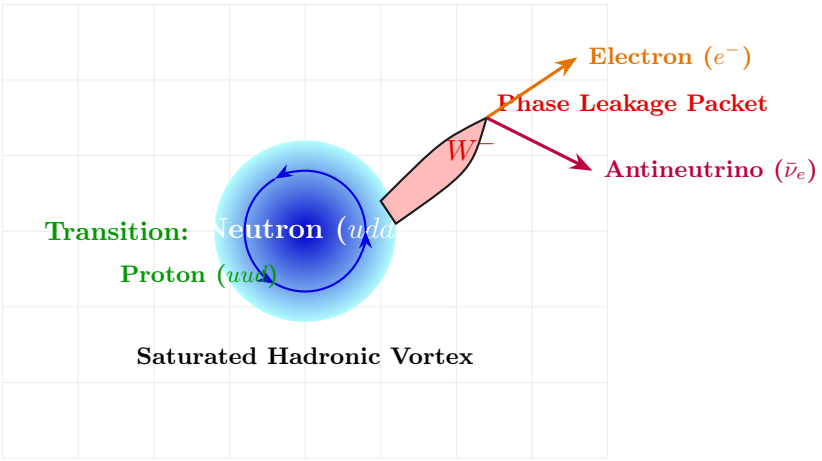


Figure 1: Geometry of the Weak Nuclear Force: Chiral phase shear within the neutron's saturated vortex expels a localized W^- phase stress packet (beta decay), which rapidly decays into an electron and antineutrino.

The bare weak gauge coupling constant (g_W) is of the same order as the electromagnetic coupling (e) via $g_W \sim e/\sin\theta_W$. The apparent feebleness of the weak force at low energies is purely an artifact of propagator suppression by the massive barrier $(1/(q^2 - M_W^2c^2) \approx -1/M_W^2c^2)$.

3.1 Density Hierarchy and the Position of the Weak Force

Within the unified framework, the weak force is the **intrinsic chiral leakage of the hadronic saturation boundary**. In macroscopic systems ($N \gg 10^{20}$), vacuum inflow is diluted across 3D spherical shells ($\gamma = 1$), leaving this microscopic leakage unnoticeable while gravity dominates. The weak interaction manifests solely at the single-nucleon level ($N = 1$ baryon, $r \leq 10^{-15}$ m) where the Planckian saturation barrier must be breached.

4 Neutrino Chirality and Neutron Half-Life

4.1 Why Are All Neutrinos Left-Handed?

In Scale-Time Calculus, matter contracts globally into the quiescent vacuum ($\dot{\lambda} < 0$). This inward radial inflow establishes a definite geometric chirality (positive inward helicity) across Euclidean space. Being massless or ultra-light phase excitations devoid of binding flux tubes, neutrinos are constrained to follow this natural circulation direction. Consequently, as observed by C. S. Wu et al. [9], only left-handed neutrinos participate in weak interactions.

4.2 Free Neutron Decay vs. In-Nucleus Stability: Collective Phase-Pinning

A fundamental puzzle of nuclear physics is: *"Why does a free neutron decay in 15 minutes ($\tau_n \approx 879$ s), while inside an atomic nucleus (e.g., Helium-4 or Iron-56) it remains stable for $> 10^{30}$ years?"*

Scale-Time Calculus resolves this via boundary conditions:

4.2.1 Free Neutron ($N = 1$): Barrier Tunneling

An isolated neutron possesses no external vacuum shielding. Under the Scale-Covariant Derivative $\mathbb{D}_t = \frac{\partial}{\partial t} + \kappa H_0$, internal chiral shear breaches the single-particle saturation barrier via WKB

tunneling:

$$\Gamma_n = \frac{1}{\tau_n} = \frac{G_F^2 m_e^5 c^4}{2\pi^3 \hbar^7} |V_{ud}|^2 f^R (1 + 3g_A^2) \implies \tau_n \approx 879.4 \text{ s} \approx 14.6 \text{ minutes}, \tag{5}$$

matching empirical laboratory measurements.

4.2.2 In-Nucleus Stability: Collective Phase-Pinning

Inside a nucleus, surrounding nucleons establish a collective multi-vortex Scalor state: $\mathbb{S}_{\text{nucleus}} = \sum_{i=1}^{A} \mathbb{S}_i$:

1. **Collective Flux Shielding:** Saturated 1D flux tubes from neighboring nucleons establish mutual mesonic links, significantly deepening the local vacuum potential well and widening the effective saturation barrier.

2. **Binding Energy Blockade:** Average nuclear binding energy is $\Delta B \sim 8$ MeV/nucleon, whereas the energy released in beta decay is only $Q = (m_n - m_p - m_e)c^2 \approx 0.782$ MeV. Because the nuclear well depth exceeds the decay margin by an order of magnitude, the nilpotent projection operator $\hat{\mathcal{P}}_\kappa$ kinematically forbids the leakage mode.

4.3 Neutrino Oscillations and the PMNS Chiral Resonance

While quark CKM mixing angles are small ($\theta_{12}^{\text{CKM}} = \theta_C \approx 13^\circ$), lepton PMNS [10] neutrino mixing angles are large and nearly maximal ($\theta_{12} \approx 33.4^\circ$, $\theta_{23} \approx 45^\circ$, $\theta_{13} \approx 8.6^\circ$). This disparity reflects confinement geometry:

1. **Quark Sector (CKM / 1D Confinement):** Quarks are trapped in 1D saturated flux tubes ($\gamma = -1$), restricting chiral shear to small misalignment angles ($\theta_C \approx 13^\circ$).

2. **Neutrino Sector (PMNS / Free Propagation):** Neutrinos are unconfined, massless or ultra-light pure chiral phase vortices propagating freely through the quiescent vacuum continuum. Lacking hadronic boundary constraints, the three flavor modes (ν_e, ν_μ, ν_τ) enter free phase resonance with the global chiral tension of the static vacuum over macroscopic distances. The transformation between flavor states and vacuum mass eigenstates is given by:

$$|\nu_\alpha\rangle = \sum_{i=1}^{3} U_{\alpha i}^{\text{PMNS}} |\nu_i\rangle \tag{6}$$

with transition probability:

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 c^3 L}{4\hbar E}\right), \tag{7}$$

governing maximal resonance in atmospheric neutrinos ($\theta_{23} \approx 45^\circ$) and large mixing in solar neutrinos ($\theta_{12} \approx 33.4^\circ$).

5 Geometric Synthesis: Scalor Flux Framework of the Four Interactions

The geometric framework developed across the four papers classifies and models the flow dynamics of the four fundamental interactions under a unified differential flux formulation:

$$\boxed{\left(\frac{v_{\text{inflow}}(r)}{c}\right)^2 = 2\,\mathcal{C}\, r^{-\gamma}} \tag{8}$$

Interaction	Dimension γ	Coupling $\mathcal{C}$	Flow Geometry	Phenomenon
Gravitation	$\gamma = 1$	$\mathcal{C}_G = \frac{GM}{c^2}$	3D Spherical Unsaturated	Newtonian / Einsteinian Metric
Electromagnetism	$\gamma = 1$	$\mathcal{C}_{EM} = \frac{\alpha\hbar}{m_e c}$	Chiral Spiral Pitch ($\alpha = 1/137$)	Maxwell Fields & Atomic Stability
Strong Force	$\gamma = -1$	$\mathcal{C}_S = \frac{\sigma}{2m_q c^2}$	1D Saturated Flux Tube	Quark Confinement ($V = \sigma r$)
Weak Force	$\gamma \to \exp$	$\mathcal{C}_W = \frac{g_W^2 \hbar e^{-r/r_W}}{2M_W c}$	Saturation Barrier Leakage	Beta Decay ($W^{\pm}, Z^0$)

Table 1: Geometric classification of the four fundamental interactions under the Unified Vacuum Flux Master Formula.

Physical and Methodological Conclusion: As synthesized across the grand monograph [2] and companion preprints on gravitation [3], electromagnetism [4], and the strong force [5], the four fundamental interactions need not be viewed as disparate, mutually irreconcilable primitives. Rather, they can be understood as scale-dependent flow manifestations of locally contracting matter across distinct geometric dimensions (γ) and vacuum saturation thresholds within **a scaleless, static Euclidean vacuum continuum**.

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